CSM
Spring 2026

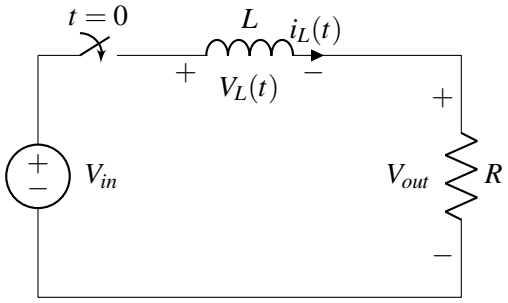
EECS 16B
Worksheet 6

1. Introduction to Inductors

Now that we are comfortable solving for the transient behavior of charging and discharging capacitors, we can move to analyzing a new circuit element: the inductor. An inductor has the physical property of *inductance*, represented by a constant L. Inductors are characterized by the following I-V relationship:

$$V_L(t) = L\frac{d}{dt}i_L(t) \tag{1}$$

Let's analyze the following LR circuit, with the initial condition $i_L(0) = 0$ with the switch open before $t = 0$:



(a) **Write out the differential equation for the current $i_L(t)$ of the inductor starting at $t = 0$ when the switch is closed.**

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(b) **Solve the differential equation for $i_L(t)$.**
HINT: given a first order differential equation $\frac{dy(t)}{dt} + \alpha y(t) = \beta$, the solution is:

$$y(t) = y(0)e^{-\alpha t} + \frac{\beta}{\alpha}(1 - e^{-\alpha t}) \tag{2}$$

(c) **What is the steady-state current through the inductor as $t \to \infty$?** Sketch a plot of the current through the inductor over time, labeling the asymptote after reaching the steady-state. This should provide you with some intuition as to the physical behavior of an inductor once an inductor is at steady state.

(d) **What is the steady-state voltage drop across the inductor as $t \to \infty$?**

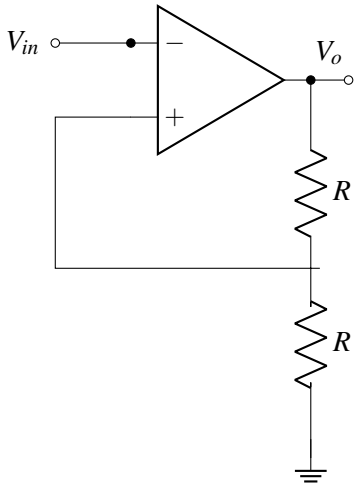
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(e) **What circuit element does the inductor act like at steady-state?**

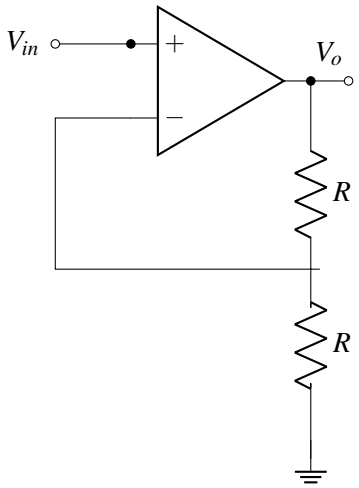
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2. OpAmps as Amplifiers

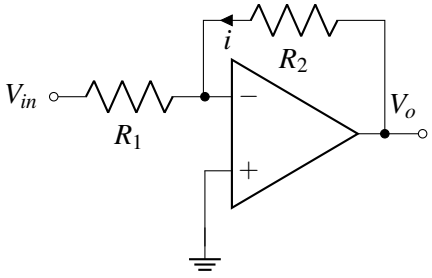
(a) In general, what is feedback? What are positive and negative feedback? And how do you introduce feedback into op-amps?



(b) Referring to the above example of an op-amp employing positive feedback, start with the assumption that $V_+ = V_-$. Now, taking into account the fact that $V_{out} = A(V_+ - V_-)$ (where A is some gain), try to raise V_{out} (by messing with V_{in}) and observe what happens. Try to lower V_{out} and observe that too.

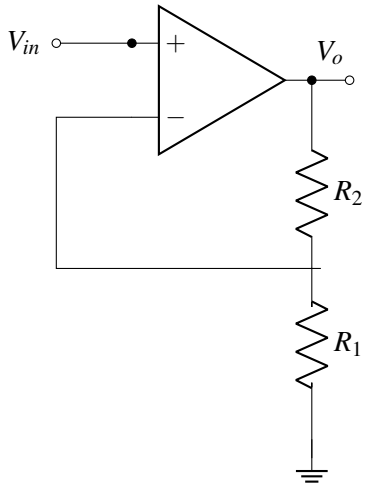


(c) Now do the same with this above example of an op-amp that uses negative feedback



(d) In terms of the input V_{in}, what is V_{out} for this circuit? What is the gain of this circuit?

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(e) Again, in terms of the input V_{in}, what is V_{out} for this circuit? What is the gain?

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3. Fun with Inductors (Challenge)

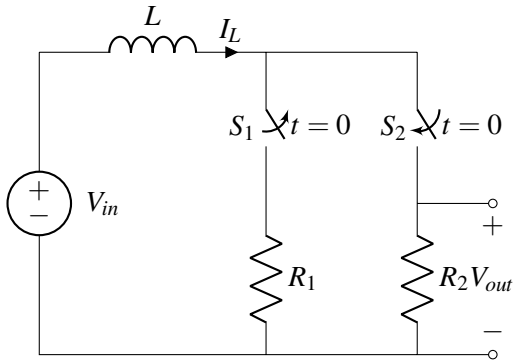


Figure 1: Circuit A

(a) Consider circuit A. Assuming that for $t < 0$, switch S_1 is on and switch S_2 is off (and both switches have been in these states indefinitely), what is $i_L(0)$?

(b) Now let's assume that for $t \geq 0$, S_1 is off and S_2 is on. Solve for $V_{out}(t)$ for $t \geq 0$.

(c) If $V_{in} = 1V$, $L = 1nH$, $R_1 = 1k\Omega$, and $R_2 = 10k\Omega$, what is the maximum value of $V_{out}(t)$ for $t \geq 0$?

(d) In general, if we want $\max V_{out}(t)$ to be greater than V_{in}, what relationship needs to be maintained between the values of R_1 and R_2?

(e) Now assume that at time $t = t_1$, switch S_2 was turned off, and switch S_1 was turned back on. Solve for $i_L(t)$ for $t > t_1$. If $R_2 > R_1$, how does this $i_L(t)$ for $t > t_1$ compare with the initial condition $i_L(0)$ you found in part (a)?

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